

TCV- ϕ VI: critical comparison, risk map, and falsification windows

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This sixth paper provides a critical comparison layer for the TCV- ϕ program. Our objective is to collect the falsifiable diagnostics obtained in Papers III–V, organize them into explicit exclusion statements, and define a risk map for the next data-level tests. The paper does not introduce new fundamental ingredients; it focuses on reproducibility, comparability, and scientific failure modes. We provide a compact quantitative matrix (primordial bridge, black-hole branch diagnostics, slow-rotation template shifts, and CC- Φ_1 microstructure indicators) together with explicit criteria under which each current interpretation would be ruled out.

I. INTRODUCTION

Papers I–V established the EFT baseline, cosmological/strong-field pipelines, and first quantum microstructure diagnostics. Paper VI consolidates these outputs into a strict falsification-oriented framework.

II. CANONICAL BASELINE AND COMPARISON PHILOSOPHY

We keep the same canonical TCV- ϕ definitions as in Papers I–V. The comparison strategy is conservative: same conventions, explicit status labels, and source-linked diagnostics.

III. CROSS-PAPER COMPARISON MATRIX

The script `compute_series_comparison.py` aggregates key artifacts from Papers III–V into `series_comparison.json`. Table I summarizes the current numerical anchors and their status labels.

IV. FALSIFICATION WINDOWS

The script `compute_falsification_table.py` produces explicit exclusion statements. A compact operational subset is shown below.

V. RISK MAP AND NEXT-TEST PRIORITIES

We rank the next tests by impact and by expected discriminatory power:

1. **Priority P1:** method-matched BH direction scans at fixed priors and matched integrators;
2. **Priority P2:** slow-rotation $\omega(r)$ closure beyond template level with explicit ODE diagnostics;
3. **Priority P3:** joint observable windows combining BH proxies and primordial constraints in a single exclusion chart.

This ordering is designed to maximize falsifiability per computational cost before broader parameter scans.

VI. STATUS LABELS AND SCIENTIFIC SCOPE

Paper VI keeps the same discipline as Papers I–V:

- derived EFT structure,
- phenomenological closures,

Sector	Quantity	Current benchmark diagnostic
Primordial bridge (Paper III)	(n_s, α_s) and CLASS bridge ratio	$n_s = 0.9643$, $\alpha_s = -1.43 \times 10^{-3}$; $P_{\text{ext}}/P_{\text{pl}} \in [0.9988, 1.0014]$
BH branch scan (Paper IV)	$\min F$ at high-density benchmark	$\rho_{\text{cc},0} = 10^{-8}$; $\min F_{\text{out}} \simeq 1.95 \times 10^{-3}$, $\min F_{\text{in}} \simeq 6.56 \times 10^{-1}$
Slow-rotation template (Paper IV)	Photon-ring proxy shift	$\Delta r_{\text{ph}}/r_{\text{ph}}^{\text{Sch}} \simeq -9.1\%$ in current template branch
CC- Φ_1 microstructure (Paper V)	Mode hierarchy and coherence response	$\Xi_2/\Xi_1 \simeq 3.72$ (equivalently $\omega_2/\omega_1 \simeq 1.93$); $L_c(R)/L_c(0) \in [0.5, 1.5]$

TABLE I. Cross-paper quantitative matrix assembled from `series_comparison.json`.

ID	Sector	Falsification statement (compact form)	Exclusion tr
F1	BH direction scan	High-density branch dependence must survive method-matched scans in the near-horizon regime.	If branch sp
F2	Slow-rotation template	Photon-ring proxy shift must remain non-zero after ODE-level closure.	If ODE-leve
F3	Primordial bridge	External-spectrum CLASS bridge must remain consistent over validated linear scales.	If improved
F4	CC- Φ_1 microstructure	Microstructure window must preserve finite coherence scales and stable low-mode hierarchy.	If refined mi

TABLE II. Operational falsification windows tracked in Paper VI.

- exploratory templates.

Its role is methodological: to maximize testability and minimize ambiguity before expanded data-level confrontations.

VII. CONCLUSION

Paper VI provides a reproducible comparison and falsification framework for the TCV- ϕ series. It defines explicit failure modes, converts cross-paper diagnostics into operational exclusion rules, and prepares the transition from internally consistent pipelines to high-stakes discriminating tests. In this sense, Paper VI acts as the methodological bridge between the current setup phase and the next targeted strong-gravity stress tests.

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- [1] C. Lecroq, “TCV- ϕ I,” preprint (2026).
 - [2] C. Lecroq, “TCV- ϕ II,” preprint (2026).
 - [3] C. Lecroq, “TCV- ϕ III,” preprint (2026).
 - [4] C. Lecroq, “TCV- ϕ IV,” preprint (2026).
 - [5] C. Lecroq, “TCV- ϕ V,” preprint (2026).